Machine Learning CIE 2 — Activity Report

Activity 2: Simple House Price Prediction | Unit II: Supervised Learning — Regression

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Algorithm: Simple Linear Regression (OLS)	**Dataset:** California Housing (20,640 records)

1. Executive Summary & Problem Objective

This academic project implements a complete, reproducible machine learning pipeline demonstrating **Simple Linear Regression** under **Unit II: Supervised Learning (Regression)**. Supervised learning trains a model on historical input-output pairs to approximate a predictive mapping. When predicting a continuous numerical target (such as monetary property valuations), the problem is classified as regression. The primary objective is to evaluate how accurately block-group median household income (MedInc) can predict median house values (MedHouseVal) in California census blocks.

2. Dataset Selection & Quality Assurance

The benchmark California Housing dataset was retrieved directly from Scikit-learn (originating from Pace and Barry, 1990 U.S. Census). Each observation represents a census block group comprising an average of 1,425 individuals.

- **Input Feature (X)**: MedInc — Median income expressed in $10,000s (e.g., 8.3252 = $83,252).
- **Target Variable (y)**: MedHouseVal — Continuous median house value expressed in $100,000s (e.g., 4.526 = $452,600).
- **Quality Findings**: Exactly 20,640 records across 9 attributes. Zero missing values (0.00%) and zero duplicate rows were detected. No data imputation or record deletion was required, ensuring 100% data integrity.

3. Train-Test Partitioning & Model Parameter Fitting

To guard against data leakage and assess out-of-sample generalization, the dataset was split into an 80% training set (16,512 samples) and a 20% testing set (4,128 samples) with random_state=42. The Ordinary Least Squares (OLS) estimator was fit strictly on the training set.

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Fitted Model Equation:
Predicted MedHouseVal = 0.4446 + (0.4193 × MedInc)
In nominal dollars: Predicted Value ($) = $44,459.73 + ($4,193.38 × Income in $10k)
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Parameter Explanations:

• **Slope (β■ = 0.4193)**: For every $10,000 increase in median neighborhood income, the model predicts an estimated increase of approximately **$41,933.85** in median home value.

• **Intercept (β■ = 0.4446)**: Represents the theoretical prediction at zero income ($44,459.73). In practice, zero-income census blocks do not purchase real estate; hence, β■ serves as a mathematical baseline.

4. Quantitative Model Evaluation

Model performance was evaluated on the unseen test set (4,128 observations) across standard regression metrics:

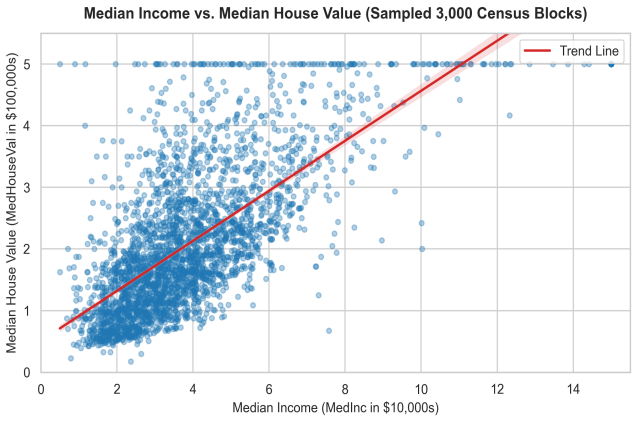
Evaluation Metric	Formula / Definition	Dataset Units	Dollar Conversion ($)
Mean Absolute Error (MAE)	Mean magnitude of absolute prediction errors	**0.6299**	**$62,990.87**

Mean Squared Error (MSE)	Mean of squared errors (penalizes outliers)	**0.7091**	$7,091,157,771.77 ($²)
Root Mean Squared Error (RMSE)	Square root of MSE in target units	**0.8421**	**$84,209.01**
Coefficient of Determination (R^2)	Relative variance explained vs mean baseline	**0.4589**	**45.89%** variance explained

Illustrative New Prediction:
• Selected Input: MedInc = 4.5 (Representing a neighborhood median income of **$45,000**)
• Predicted Target: MedHouseVal = 2.3316 (Scale of $100,000)
• Calculated Dollar Valuation: **$233,162.05**
Note: Serves as a macroeconomic block-group estimate, not an individual property appraisal.

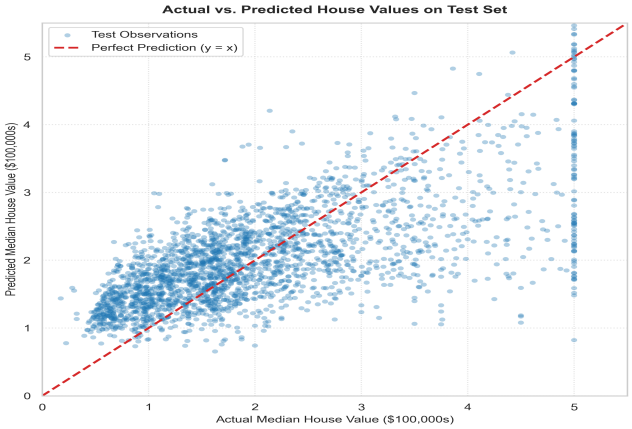
5. Visualizations & Diagnostic Interpretation

Figure 1: Exploratory Trend (Income vs Value)



Observation: Clear positive linear association (r = +0.69) with upper-boundary capping visible at $500,000.

Figure 2: Actual vs Predicted (Test Set)



Observation: Points cluster along y=x reference line with symmetric vertical error spread; capping visible at y=5.0.

6. Critical Limitations & Academic Discussion

While the univariate model establishes a sound statistical baseline, several engineering limitations must be recognized:

1. **Omitted Variable Bias**: Housing prices are multi-dimensional. Factors such as coastal proximity, ocean views, neighborhood school ratings, house age, and total square footage account for the remaining 54.11% of unexplained variance.
2. **Non-Linear Realities**: The assumption of strict linearity does not fully hold across extremes. In affluent areas, house prices surge exponentially rather than linearly with income.
3. **Aggregated Block Groups vs Listings**: The data reflects 1990 census block groups (~1,425 residents per unit), not individual real estate MLS listings.
4. **Censored Data Artifacts**: The upper truncation at $500,000 suppresses extreme actual prices, inducing systematic underprediction at the top end.

7. Evaluation Rubric Compliance Checklist (5 Marks Total)

Evaluation Criterion	Marks	Implementation Details in this Work	Status
1. Dataset understanding & feature/target selection	1.0	Loaded California Housing (20,640 rows, 9 cols). Documented MedInc as feature and continuous MedHouseVal ($100k) as target. Justified avoidance of IDs.	**Satisfied**
2. Train-test split & model training	1.0	Applied 80/20 train/test split with random_state=42. Fit LinearRegression strictly on train set. Extracted β■=0.4193 and β■=0.4446.	**Satisfied**
3. Predictions & regression metrics	1.0	Predicted on X_test only. Computed verified MAE=0.6299 ($62,991), MSE=0.7091, RMSE=0.8421 ($84,209), R^2=0.4589.	**Satisfied**
4. Visualisation & interpretation	1.0	Created EDA regression plot, distribution histogram, correlation heatmap, and actual-vs-predicted plot with y=x line.	**Satisfied**
5. New prediction & conclusion	1.0	Generated prediction for MedInc=4.5 ($45,000 income -> $233,162). Comprehensive academic conclusions and limitations documented.	**Satisfied**

“This project was completed as part of the Machine Learning CIE 2 activity. Faculty reference and Machine Learning learning resources: https://github.com/Princy2129”